Preparing Bell States from $|00\rangle$ with Quantum Circuits

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FYS5419/9419, examples on how to set up gates

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Goal: Create Bell states starting from $|00\rangle$

We start with the two-qubit computational basis state

$$|\psi_0\rangle = |00\rangle \,.$$

Using a small set of gates we can prepare the four Bell states:

$$|\Phi^{\pm}\rangle = \frac{|00\rangle \pm |11\rangle}{\sqrt{2}}, \qquad |\Psi^{\pm}\rangle = \frac{|01\rangle \pm |10\rangle}{\sqrt{2}}.$$

Core idea:

- Use a Hadamard on qubit 0 to create a superposition.
- Use an entangling gate (CNOT) to correlate qubit 1 with qubit 0.
- Apply simple single-qubit gates (X, Z) to obtain the other Bell states.

Building blocks: single- and two-qubit gates

Hadamard gate on one qubit:

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \qquad H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

CNOT with control qubit 0 and target qubit 1:

$$\mathrm{CNOT}\,|c, t\rangle = |c,\, t \oplus c\rangle\,.$$

Thus, if $c = 0$ nothing happens; if $c = 1$, the target bit flips.

We also use Pauli gates:

$$X|0\rangle = |1\rangle\,,\ X|1\rangle = |0\rangle\,, \qquad Z|0\rangle = |0\rangle\,,\ Z|1\rangle = -|1\rangle\,.$$

Circuit for $|\Phi^+\rangle$ from $|00\rangle$

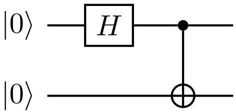
Start with $|00\rangle$.

Step 1: Apply H on qubit 0:

$$(H \otimes I)\,|00\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle + |10\rangle\right).$$

Step 2: Apply $\mathrm{CNOT}_{0\to 1}$:

$$\mathrm{CNOT}\,\frac{|00\rangle + |10\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right) = |\Phi^+\rangle\,.$$



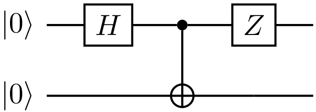
Circuit for $|\Phi^-\rangle$ from $|00\rangle$

Use the same entangling circuit as for $|\Phi^+\rangle$, then add a Z phase flip.

Since

$$(I \otimes Z)\,|\Phi^+\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle - |11\rangle\right) = |\Phi^-\rangle\,,$$

we can implement $|\Phi^-\rangle$ by applying Z to either qubit (global phase conventions aside).



Equivalent variant (apply Z on the second qubit):

$$(I \otimes Z)\,|\Phi^+\rangle = |\Phi^-\rangle\,.$$

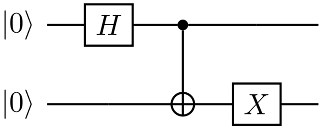
Circuit for $|\Psi^+\rangle$ from $|00\rangle$

One convenient route:

$$|\Psi^+\rangle = (I \otimes X)\,|\Phi^+\rangle\,.$$

Thus: build $|\Phi^+\rangle$ and then flip the second qubit with X:

$$(I \otimes X)\frac{|00\rangle + |11\rangle}{\sqrt{2}} = \frac{|01\rangle + |10\rangle}{\sqrt{2}} = |\Psi^+\rangle\,.$$



(You can also apply X before the CNOT on the target qubit; the above is the clearest construction.)

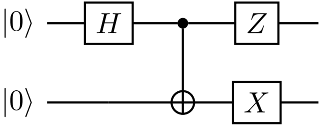
Circuit for $|\Psi^-\rangle$ from $|00\rangle$

A simple construction uses

$$|\Psi^-\rangle = (I \otimes X)\,|\Phi^-\rangle \quad \text{or} \quad |\Psi^-\rangle = (I \otimes Z)(I \otimes X)\,|\Phi^+\rangle\,.$$

Starting from $|\Phi^+\rangle$:

$$|\Phi^+\rangle \xrightarrow{X \text{ on qubit } 1} |\Psi^+\rangle \xrightarrow{Z \text{ on qubit 0 (or 1)}} |\Psi^-\rangle\,.$$



Up to a global phase, applying Z on either qubit produces the relative minus sign between $|01\rangle$ and $|10\rangle$.

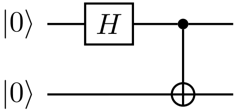
Summary: One entangling core circuit + local Pauli gates

Core entangling circuit prepares $|\Phi^+\rangle$:

$$|00\rangle \xrightarrow{H\otimes I} \frac{|00\rangle + |10\rangle}{\sqrt{2}} \xrightarrow{\mathrm{CNOT}_{0\to 1}} |\Phi^+\rangle \,.$$

Then obtain the other Bell states by local operations:

$$|\Phi^-\rangle = (Z \otimes I)\,|\Phi^+\rangle \,, \qquad |\Psi^+\rangle = (I \otimes X)\,|\Phi^+\rangle \,, \qquad |\Psi^-\rangle = (Z \otimes I)(I \otimes X)\,|\Phi^+\rangle \,.$$



Optional: Verify with state evolution (conceptual)

If we denote the circuit unitary by U, then

$$U|00\rangle = |\Phi^+\rangle .$$

Local Pauli operations generate the full Bell basis:

$$\{|\Phi^+\rangle, (Z\otimes I)|\Phi^+\rangle, (I\otimes X)|\Phi^+\rangle, (Z\otimes I)(I\otimes X)|\Phi^+\rangle\}.$$

In quantum information, this is summarized as:

$$\text{Bell basis} = \{(I\otimes\sigma)|\Phi^+\rangle \;:\; \sigma\in\{I,X,Z,XZ\}\}.$$